

Titanic_Arbeit

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#INTRODUCTION The sinking of the Titanic is one of the most infamous shipwrecks in history. In this challenge, we ask you to build a predictive model that answers the question: “what sorts of people were more likely to survive?” using passenger data (ie name, age, gender, socio-economic class, etc).

#DATA UNDERSTANDING AND VISUALIZATION

```
T<-read.csv("train.csv")
T2 <- read.csv("test.csv")
str(T)
```

```
## 'data.frame': 891 obs. of 12 variables:
## $ PassengerId: int 1 2 3 4 5 6 7 8 9 10 ...
## $ Survived : int 0 1 1 1 0 0 0 0 1 1 ...
## $ Pclass : int 3 1 3 1 3 3 1 3 3 2 ...
## $ Name : chr "Braund, Mr. Owen Harris" "Cumings, Mrs. John Bradley (Florence Briggs Thayer)"
## $ Sex : chr "male" "female" "female" "female" ...
## $ Age : num 22 38 26 35 35 NA 54 2 27 14 ...
## $ SibSp : int 1 1 0 1 0 0 0 3 0 1 ...
## $ Parch : int 0 0 0 0 0 0 0 1 2 0 ...
## $ Ticket : chr "A/5 21171" "PC 17599" "STON/O2. 3101282" "113803" ...
## $ Fare : num 7.25 71.28 7.92 53.1 8.05 ...
## $ Cabin : chr "" "C85" "" "C123" ...
## $ Embarked : chr "S" "C" "S" "S" ...
```

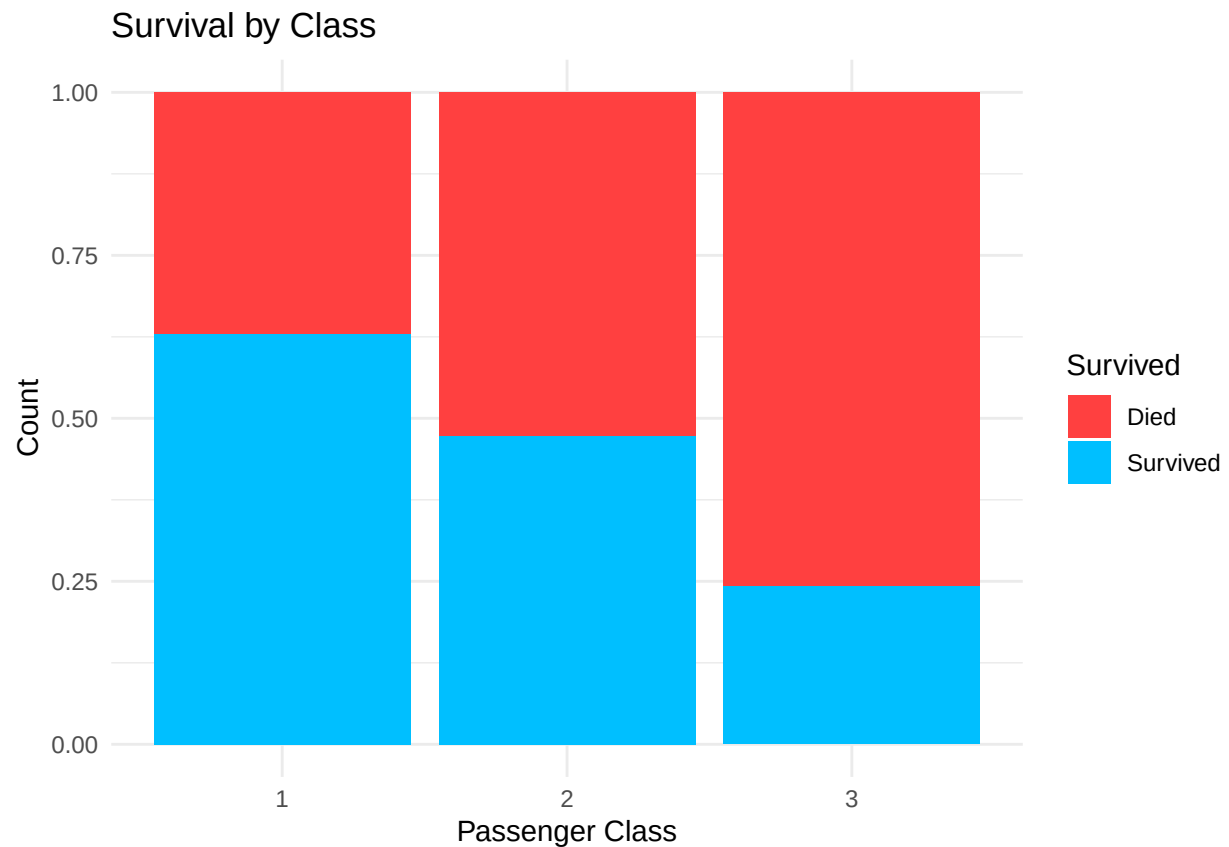
```
str(T2)
```

```
## 'data.frame': 418 obs. of 11 variables:
## $ PassengerId: int 892 893 894 895 896 897 898 899 900 901 ...
## $ Pclass : int 3 3 2 3 3 3 3 2 3 3 ...
## $ Name : chr "Kelly, Mr. James" "Wilkes, Mrs. James (Ellen Needs)" "Myles, Mr. Thomas Francis"
## $ Sex : chr "male" "female" "male" "male" ...
## $ Age : num 34.5 47 62 27 22 14 30 26 18 21 ...
## $ SibSp : int 0 1 0 0 1 0 0 1 0 2 ...
## $ Parch : int 0 0 0 0 1 0 0 1 0 0 ...
## $ Ticket : chr "330911" "363272" "240276" "315154" ...
## $ Fare : num 7.83 7 9.69 8.66 12.29 ...
## $ Cabin : chr "" "" "" "" ...
## $ Embarked : chr "Q" "S" "Q" "S" ...
```

```
summary(T)
```

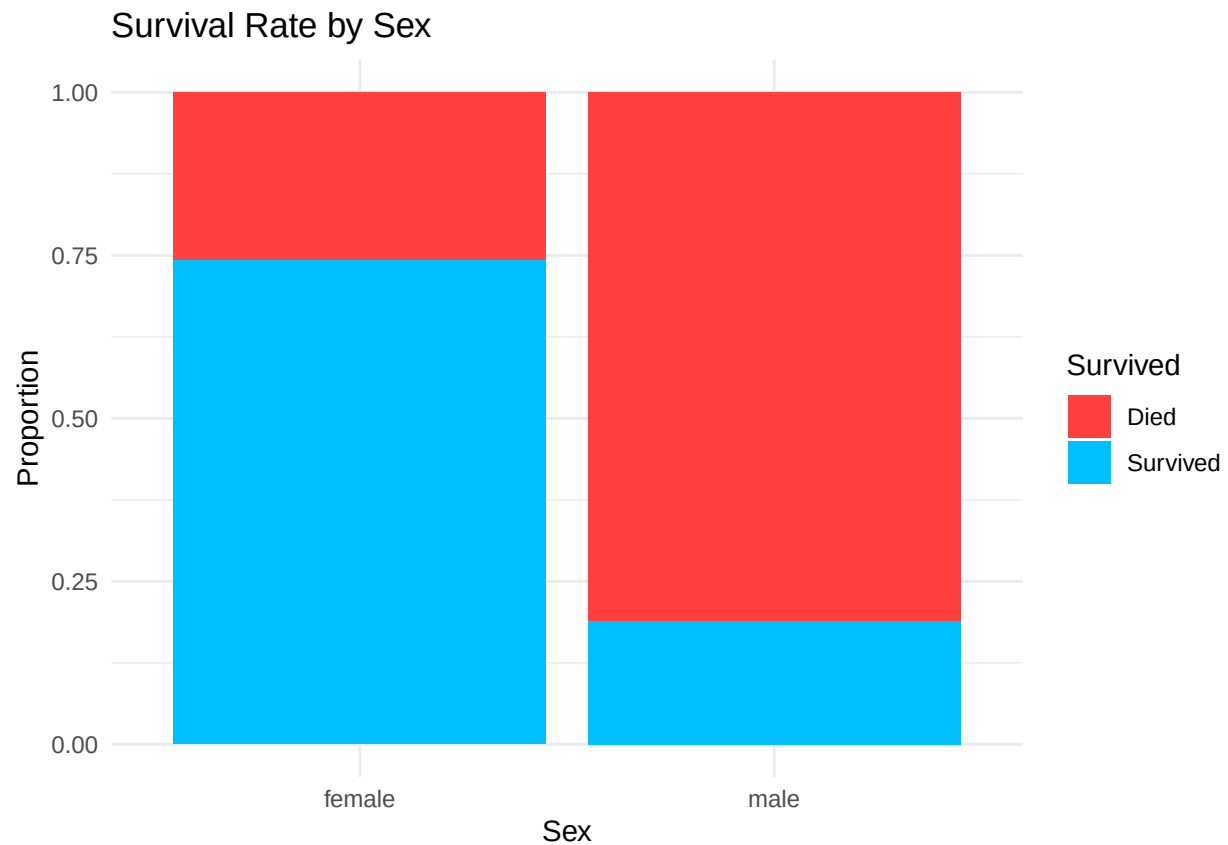
```
## PassengerId      Survived      Pclass      Name
## Min.   : 1.0      Min.   :0.0000   Min.   :1.000   Length:891
## 1st Qu.:223.5     1st Qu.:0.0000   1st Qu.:2.000   Class :character
## Median :446.0     Median :0.0000   Median :3.000   Mode  :character
## Mean   :446.0     Mean   :0.3838   Mean   :2.309
## 3rd Qu.:668.5     3rd Qu.:1.0000   3rd Qu.:3.000
## Max.   :891.0     Max.   :1.0000   Max.   :3.000
##
## Sex              Age              SibSp              Parch
## Length:891      Min.   : 0.42   Min.   :0.000   Min.   :0.0000
## Class :character 1st Qu.:20.12   1st Qu.:0.000   1st Qu.:0.0000
## Mode  :character Median :28.00   Median :0.000   Median :0.0000
##                      Mean   :29.70   Mean   :0.523   Mean   :0.3816
##                      3rd Qu.:38.00   3rd Qu.:1.000   3rd Qu.:0.0000
##                      Max.   :80.00   Max.   :8.000   Max.   :6.0000
##                      NA's   :177
## Ticket          Fare              Cabin              Embarked
## Length:891      Min.   : 0.00   Length:891     Length:891
## Class :character 1st Qu.: 7.91   Class :character Class :character
## Mode  :character Median :14.45   Mode  :character Mode  :character
##                      Mean   :32.20
##                      3rd Qu.:31.00
##                      Max.   :512.33
##
```

```
ggplot(T, aes(x = factor(Pclass), fill = factor(Survived))) +
  geom_bar(position = "fill") +
  labs(
    title = "Survival by Class",
    x = "Passenger Class",
    y = "Count",
    fill = "Survived"
  ) +
  scale_fill_manual(values = c("brown1", "deepskyblue"),
                    labels = c("Died", "Survived")) +
  theme_minimal()
```



#People in third class had a significantly lower survival probability than those in other classes. Around

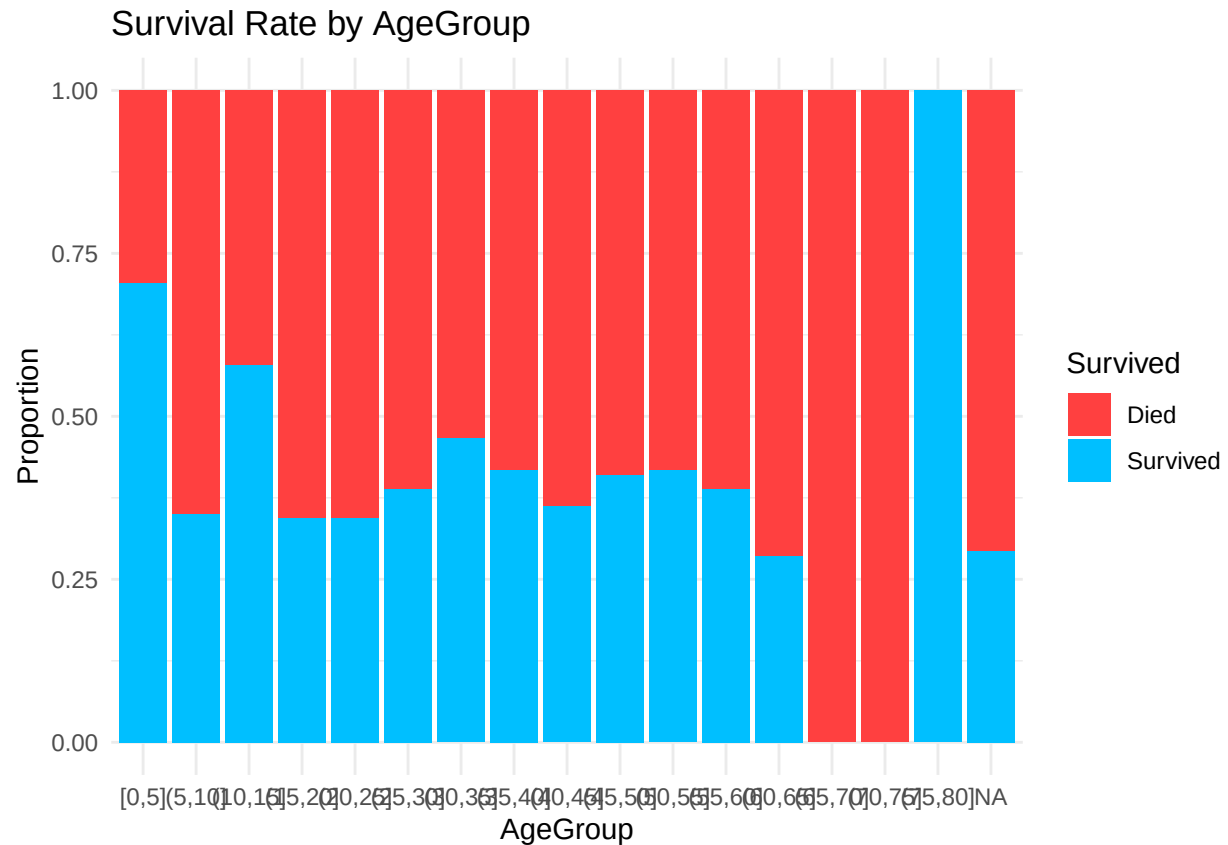
```
ggplot(T, aes(x = Sex, fill = factor(Survived))) +
  geom_bar(position = "fill") +
  labs(
    title = "Survival Rate by Sex",
    x = "Sex",
    y = "Proportion",
    fill = "Survived"
  ) +
  scale_fill_manual(values = c("brown1", "deepskyblue"),
    labels = c("Died", "Survived")) +
  theme_minimal()
```



```
T$AgeGroup <- cut(
  T$Age,
  breaks = seq(0, 80, by = 5),
  include.lowest = TRUE
)
table(T$AgeGroup)
```

```
##
##  [0,5]  (5,10] (10,15] (15,20] (20,25] (25,30] (30,35] (35,40] (40,45] (45,50]
##      44      20      19      96     122     108      88      67      47      39
## (50,55] (55,60] (60,65] (65,70] (70,75] (75,80]
##      24      18      14       3       4       1
```

```
ggplot(T, aes(x = AgeGroup, fill = factor(Survived))) +
  geom_bar(position = "fill") +
  labs(
    title = "Survival Rate by AgeGroup",
    x = "AgeGroup",
    y = "Proportion",
    fill = "Survived"
  ) +
  scale_fill_manual(values = c("brown1", "deepskyblue"),
                    labels = c("Died", "Survived")) +
  theme_minimal()
```



MISSING VALUES, AMPUTATION AND FEATURE ENGINEERING

```
colSums(is.na(T))
```

```
## PassengerId    Survived    Pclass      Name      Sex      Age
##           0           0           0           0           0      177
##      SibSp      Parch      Ticket     Fare      Cabin Embarked
##           0           0           0           0           0         0
##   AgeGroup
##        177
```

```
round(colSums(is.na(T)) / nrow(T) * 100, 2)
```

```
## PassengerId    Survived    Pclass      Name      Sex      Age
##         0.00         0.00         0.00         0.00         0.00    19.87
##      SibSp      Parch      Ticket     Fare      Cabin Embarked
##         0.00         0.00         0.00         0.00         0.00         0.00
##   AgeGroup
##        19.87
```

#only Age had 177 missing values. Since 17% of the values are missing, I can replace them with the mean

```
medianage<-median(T$Age,na.rm=TRUE)
meanmedian<-mean(T$Age, na.rm = TRUE)
T$Age[is.na(T$Age)]<-medianage
```

```
#Cabin
sum(T$Cabin=="") #687 of 891
```

```
## [1] 687
```

```
T$CabinKnown <- ifelse(T$Cabin == "", 0, 1)
```

```
#In the cabin section, there are a lot of missing values right now.
#I don't know why #they haven't been entered (I'll look into it later).
#For now, I've assigned 0 to the missing values and 1 to the opposite.
#I'll create a model that distinguishes between cabins and non-cabins.
#Because I think the rate of missing values is too high.
```

```
#### Test values
colSums(is.na(T2))
```

```
## PassengerId      Pclass      Name      Sex      Age      SibSp
##           0           0           0           0      86           0
##      Parch      Ticket      Fare      Cabin      Embarked
##           0           0           1           0           0
```

```
round(colSums(is.na(T2)) / nrow(T2) * 100, 2)
```

```
## PassengerId      Pclass      Name      Sex      Age      SibSp
##          0.00      0.00      0.00      0.00     20.57      0.00
##      Parch      Ticket      Fare      Cabin      Embarked
##          0.00      0.00      0.24      0.00      0.00
```

```
# only Age had 177 missing values.
# Since 17% of the values are missing, I can replace them with the mean or the median.
```

```
medianage<-median(T2$Age,na.rm=TRUE)
meanmedian<-mean(T2$Age, na.rm = TRUE)
T2$Age[is.na(T2$Age)]<-medianage
```

```
#Cabin
sum(T2$Cabin=="") #687 of 891
```

```
## [1] 327
```

```
T2$CabinKnown <- ifelse(T2$Cabin == "", 0, 1)
```

```
#The encoding of sex and Embarkation to see the correlation with another variables (likesurvived, pclass.
#sex male(0), female(1)
T$Sexnew<- ifelse(T$Sex=="male",0,1)
# Port of Embarkation C = Cherbourg(1), Q = Queenstown(2), S = Southampton(0)
T$Embarkednew<- ifelse(T$Embarked=="S",0,
```

```

        ifelse(T$Embarked=="C",1,2))

#####
T2$Sexnew<- ifelse(T2$Sex=="male",0,1)
# Port of Embarkation C = Cherbourg(1), Q = Queenstown(2), S = Southampton(0)

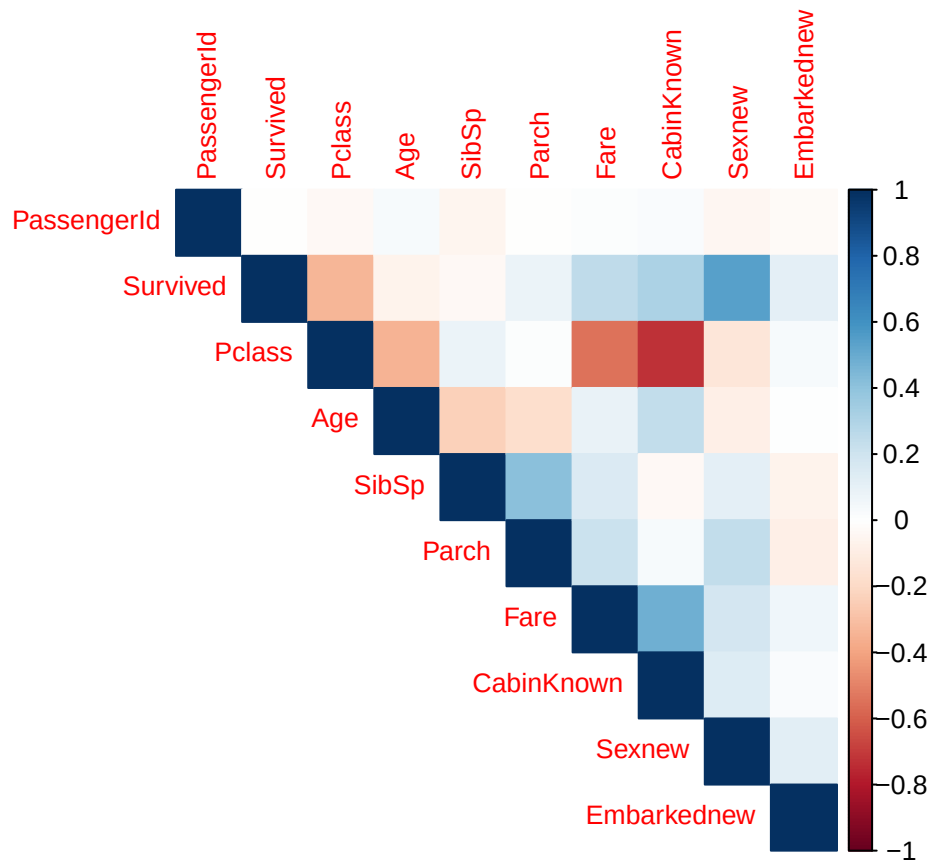
T2$Embarkednew<- ifelse(T2$Embarked=="S",0,
        ifelse(T2$Embarked=="C",1,2))

#####
numeric_vars <- T[, sapply(T, is.numeric)]
cor_matrix <- round(cor(numeric_vars, use = "complete.obs"),3)
cor_matrix

##      PassengerId Survived Pclass   Age  SibSp  Parch   Fare CabinKnown
## PassengerId      1.000  -0.005 -0.035  0.034 -0.058 -0.002  0.013      0.020
## Survived        -0.005   1.000 -0.338 -0.065 -0.035  0.082  0.257      0.317
## Pclass          -0.035  -0.338  1.000 -0.340  0.083  0.018 -0.549     -0.726
## Age             0.034  -0.065 -0.340  1.000 -0.233 -0.172  0.097      0.240
## SibSp           -0.058  -0.035  0.083 -0.233  1.000  0.415  0.160     -0.040
## Parch           -0.002   0.082  0.018 -0.172  0.415  1.000  0.216      0.037
## Fare            0.013   0.257 -0.549  0.097  0.160  0.216  1.000      0.482
## CabinKnown      0.020   0.317 -0.726  0.240 -0.040  0.037  0.482      1.000
## Sexnew          -0.043   0.543 -0.132 -0.081  0.115  0.245  0.182      0.140
## Embarkednew     -0.030   0.115  0.034  0.002 -0.063 -0.081  0.068      0.027
##      Sexnew Embarkednew
## PassengerId -0.043      -0.030
## Survived     0.543      0.115
## Pclass       -0.132      0.034
## Age          -0.081      0.002
## SibSp        0.115     -0.063
## Parch        0.245     -0.081
## Fare         0.182      0.068
## CabinKnown   0.140      0.027
## Sexnew       1.000      0.125
## Embarkednew  0.125      1.000

corrplot(cor_matrix, method = "color", type = "upper", tl.cex = 0.8)

```



```
#work on the another character
T$title<- sub(".*, (.*?)\\..*", "\\1",T$Name)
round(prop.table(table(T$title)),3)
```

```
##
##      Capt      Col      Don      Dr      Jonkheer      Lady
##      0.001      0.002      0.001      0.008      0.001      0.001
##      Major      Master      Miss      Mlle      Mme      Mr
##      0.002      0.045      0.204      0.002      0.001      0.580
##      Mrs      Ms      Rev      Sir the Countess
##      0.140      0.001      0.007      0.001      0.001
```

```
#the important title are Mr, Miss(Miss,Mlle,Mrs,mme), Master and rare(capt,col,Don,Dr,Jonkheer, Major,s
T$title_group <- ifelse(T$title %in% c("Mr"), "Mr",
  ifelse(T$title %in% c("Miss","Mlle","Ms"), "Miss",
    ifelse(T$title %in% c("Mrs","Mme"), "Mrs",
      ifelse(T$title == "Master", "Master", "Rare"))))
```

```
#test values
T2$title<- sub(".*, (.*?)\\..*", "\\1",T2$Name)
round(prop.table(table(T2$title)),3)
```

```
##
##      Col      Dona      Dr Master      Miss      Mr      Mrs      Ms      Rev
```



```
## 0.005 0.002 0.002 0.050 0.187 0.574 0.172 0.002 0.005
```

```
#the important title are Mr, Miss(Miss,Mlle,Mrs,mme), Master and rare(capt,col,Don,Dr,Jonkheer, Major,s
```

```
T2$title_group <- ifelse(T2$title %in% c("Mr"), "Mr",
  ifelse(T2$title %in% c("Miss","Mlle","Ms"), "Miss",
  ifelse(T2$title %in% c("Mrs","Mme"), "Mrs",
  ifelse(T2$title == "Master", "Master", "Rare"))))
```

Logistic regression

```
model <- glm(Survived ~ Pclass + Sexnew + Age + Fare+Embarkednew +CabinKnown,
  data = T,
  family = binomial)

pred <- predict(model, newdata = T, type = "response")
pred_class <- ifelse(pred > 0.57, 1, 0) #1= survived 0=died

train_eval<- table(Predicted = pred_class, Actual = T$Survived)
# train evaluation

TP1 <- train_eval["1","1"]
TN1 <- train_eval["0","0"]
FP1 <- train_eval["1","0"]
FN1 <- train_eval["0","1"]
Accuracy <- (TP1+TN1)/(TP1+FN1+TN1+FP1)
ConfusionMatrix<- confusionMatrix(as.factor(pred_class), as.factor(T$Survived))
print(ConfusionMatrix)
```

Confusion Matrix and Statistics

```
##
##           Reference
## Prediction  0    1
##           0 492 115
##           1  57 227
##
##           Accuracy : 0.807
##           95% CI : (0.7795, 0.8324)
##           No Information Rate : 0.6162
##           P-Value [Acc > NIR] : < 2.2e-16
##
##           Kappa : 0.5784
##
## Mcnemar's Test P-Value : 1.385e-05
##
##           Sensitivity : 0.8962
##           Specificity : 0.6637
##           Pos Pred Value : 0.8105
##           Neg Pred Value : 0.7993
##           Prevalence : 0.6162
##           Detection Rate : 0.5522
```

```
##      Detection Prevalence : 0.6813
##      Balanced Accuracy   : 0.7800
##
##      'Positive' Class    : 0
##
```

```
#Using the data from the Titanic passenger manifest (test data),
#I have compiled a list of people who might have survived based on
#my logistic regression model trained above.
#pred <- predict(model, newdata = T2, type = "response")
#pred_class <- ifelse(pred > 0.57, 1, 0)

#T2$survived <- pred_class
#submission <- T2[, c("PassengerId", "Survived")]
```